

Clinical Risk Assessment Report: Patient 99

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: High Confidence
- Absolute Predicted Risk: 38.6% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.74x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET), which elevated the patient's absolute risk by +6.4%. Biologically, this feature points to continuous, elongated bands of high-density or highly active tumor tissue.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: 1.49 [Top 7%] (+6.4%)**
Points to continuous, elongated bands of high-density or highly active tumor tissue.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.86 [Top 18%] (+5.5%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Tumor Pixel Correlation (CT) (GLCM_Correlation.CT) **Value: 2.81 [Top <1%] (+3.8%)**
Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: 1.16 [Top 4%] (+3.4%)**
Reflects the most dense solid components or calcifications within the primary tumor lesion.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 1.13 [Top 12%] (+2.8%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: 0.01 [Top 11%] (+2.7%)**
Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

3. Protective / Mitigating Factors (Reducing Risk)

Tumor Sphericity (PET) (SHAPE_Sphericity(onlyFor3DROI)).PET) **Value: -0.66 [Bottom 20%] (-1.7%)**
Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: 1.18 [Top 2%] (-1.7%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Machine Learning Feature Attribution (SHAP Decision Path)

